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Do Vision-Language Models Really See What They Say?

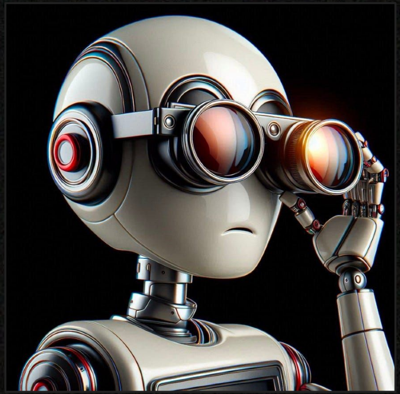
https://github.com/gaxur/MDN-C-CS

Motivation

Why I chose this topic

Vision-Language Models (VLMs) have made remarkable progress in understanding images, yet they suffer from a critical failure mode: **hallucination**. These models confidently describe visual content that is not actually present in the image.

So, I set out to systematically measure and compare hallucination rates across two state-of-the-art VLMs..



Furthermore, other projects in this course build on top of VLMs . . .

So, if the underlying model hallucinates, every system above it inherits that flaw — silently, and often without any warning, being very dangerous in some real cases :

- **Medical imaging**: Misdiagnosis risks
- **Autonomous driving**: Safety-critical decisions
- **Scientific analysis**: Data integrity and reliability

Goals

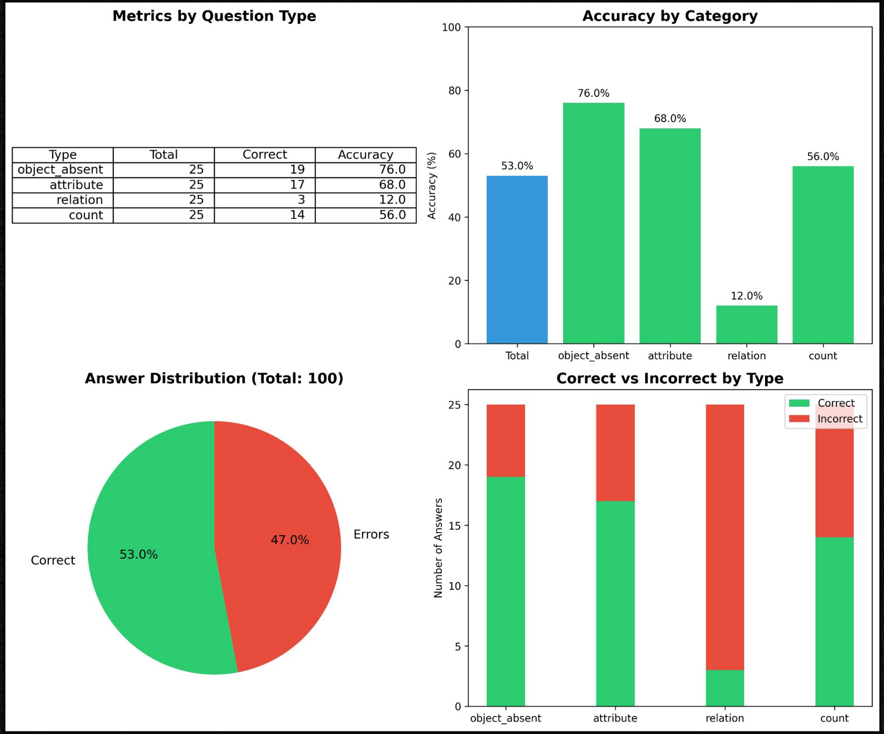
- Build a benchmark of carefully designed image-question pairs to systematically probe hallucination behavior
- Evaluate two VLMs: LLaVA-v1.5 and Qwen3-VL (optional internvl35)
- Analyze hallucination patterns across four categories:
    - Object hallucinations
    - Color hallucinations
    - Relation hallucinations
    - Counting hallucinations
- Test a mitigation strategy using stricter prompt engineering

InternVL

Qwen3-VL

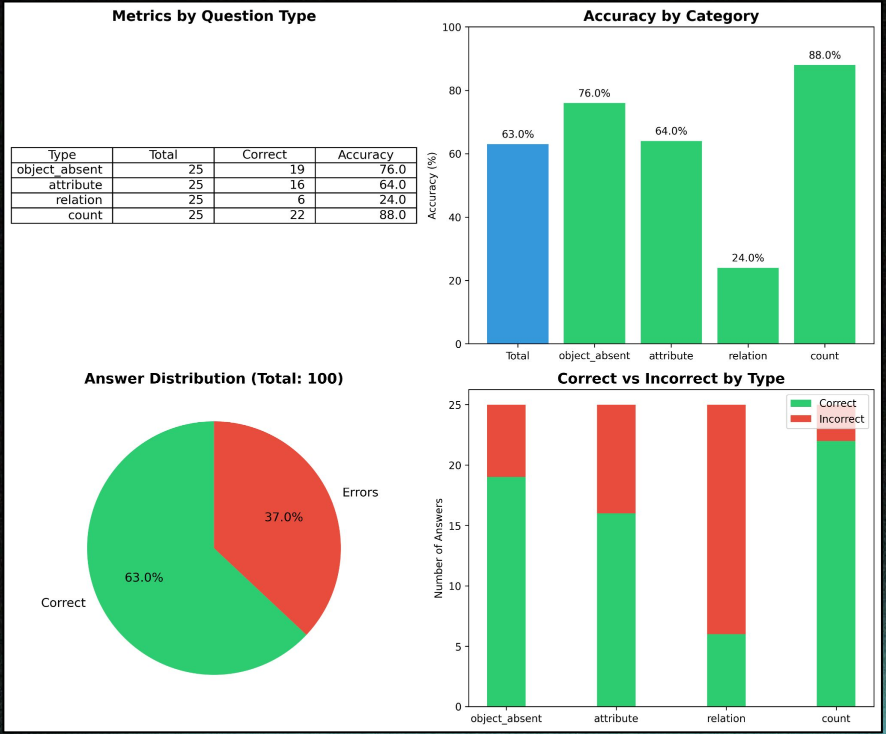
Results

reports/llava15/complete_results.png



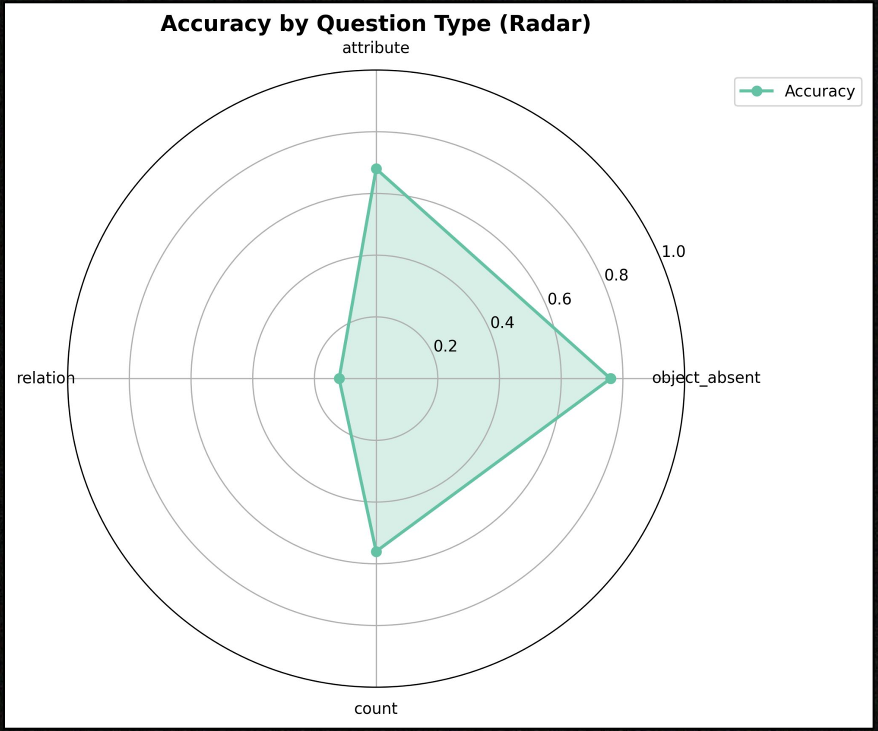
Type	Total	Correct	Accuracy
object_absent	25	19	76.0
attribute	25	17	68.0
relation	25	3	12.0
count	25	14	56.0

reports/llava15_strict/complete_results.png

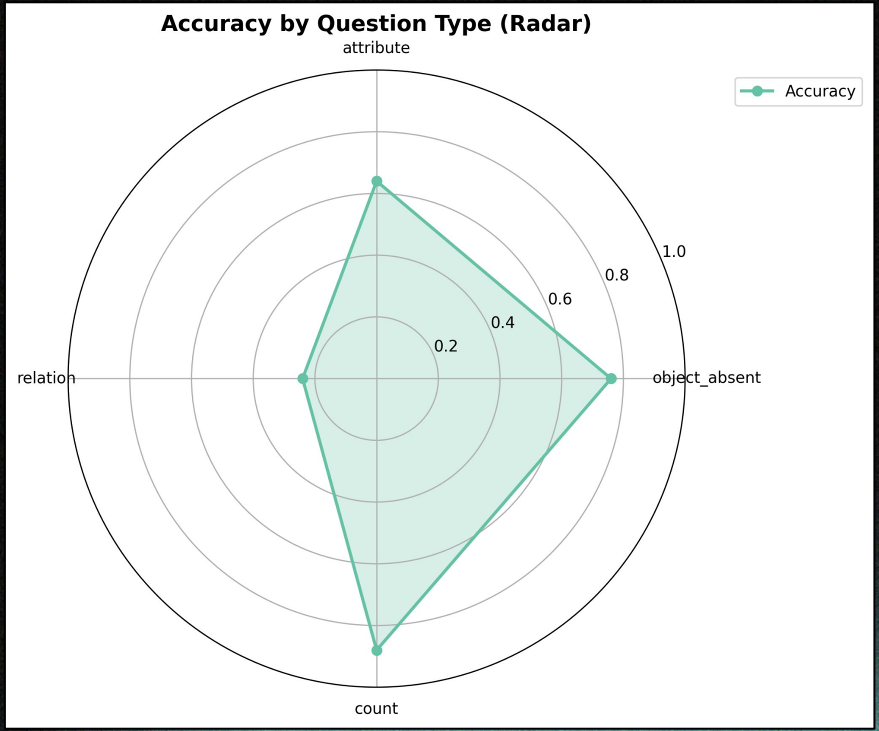


Type	Total	Correct	Accuracy
object_absent	25	19	76.0
attribute	25	16	64.0
relation	25	6	24.0
count	25	22	88.0

reports/llava15/radar_accuracy.png



reports/llava15_strict/radar_accuracy.png



Conclusions

Key Findings

LLaVA-v1.5 is more prone to hallucination
Strict prompting reduces hallucination, especially in the LLaVA-v1.5
Most critical hallucination type: Spatial Relations

Implications

VLMs require careful evaluation before deployment in safety-critical applications
Prompt engineering is an effective mitigation technique
Further research needed on robust hallucination detection

Future Work

Evaluate additional models (GPT-4 Vision)
Test other mitigation strategies (OPERA)
Apply findings to production systems